

Agentic DevOps with GitHub

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Building agents brings new challenges...

Traditional apps

Fixed rules, manual updates

Rigid, predefined steps

Infrastructure-heavy, platform-specific code

Coding & backend

Logic

Workflow

Development focus

Skills

AI agents

Autonomous, adaptive reasoning

Dynamic, iterative loops

Intelligence, orchestration, adaptability

Specification-driven, orchestration

Building with agents scales the skills of every dev

Every development team—and every person on that team—brings something to the table



Competence: What they know



Capability: What they can do



Capacity: How much they can accomplish

Every role



Competence
Knowledge



Capacity
Productivity

+ Agentic AI

+ Generative AI

Developers

Senior

Junior



Capability
Skills



Create

Collaborate

Operate

the new command center...|

Agentic layer



Create

Collaborate

Operate

GitHub Copilot

the new command center...|

Agentic layer



Agent mode

Code review

Coding agent

Copilot autofix*

GitHub Copilot for Azure

SRE agent

*Requires GitHub Advanced Security

GitHub Copilot

across the software development lifecycle



Create
in the IDE



Collaborate
across the SDLC

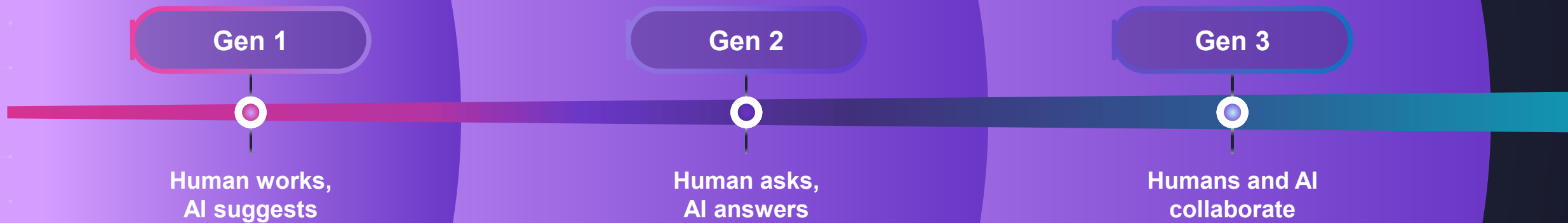


Operate
in the cloud

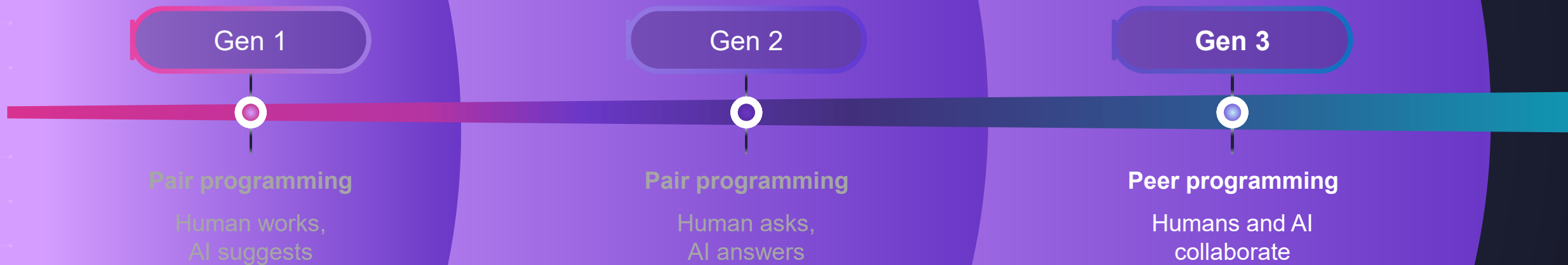


GitHub Copilot

Evolution of agentic workflows



Evolution of agentic workflows

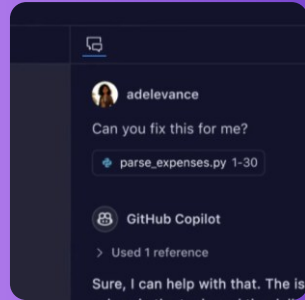


Evolution of GitHub Copilot



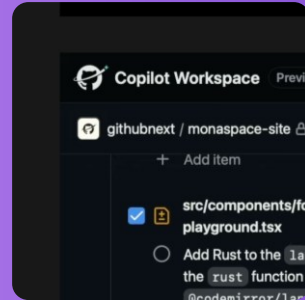
Jun 2021

Code completions



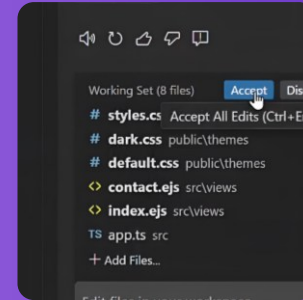
Mar 2023

Chat



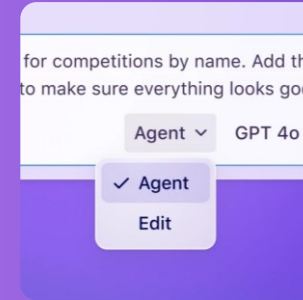
Nov 2023

Workspace



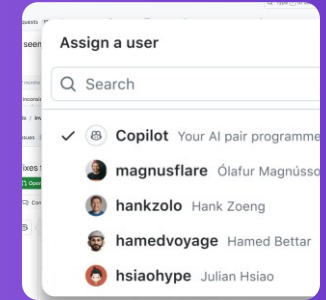
Oct 2024

Multi-file edits



Feb 2025

Agent mode



May 2025

Coding agent

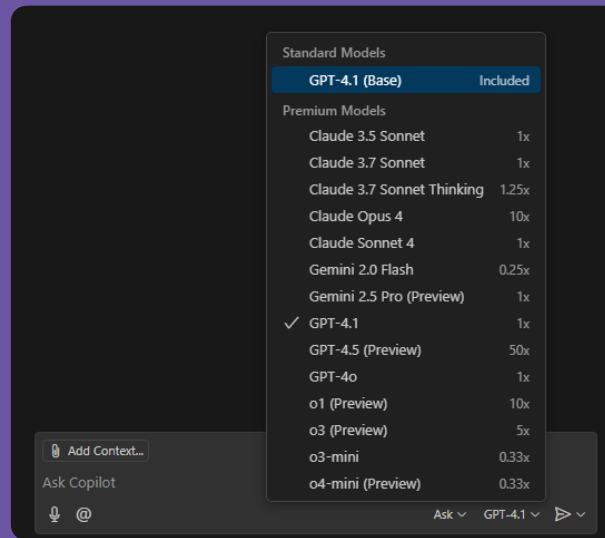
GitHub Copilot meets you where you are



Favorite IDE



Models of Choice



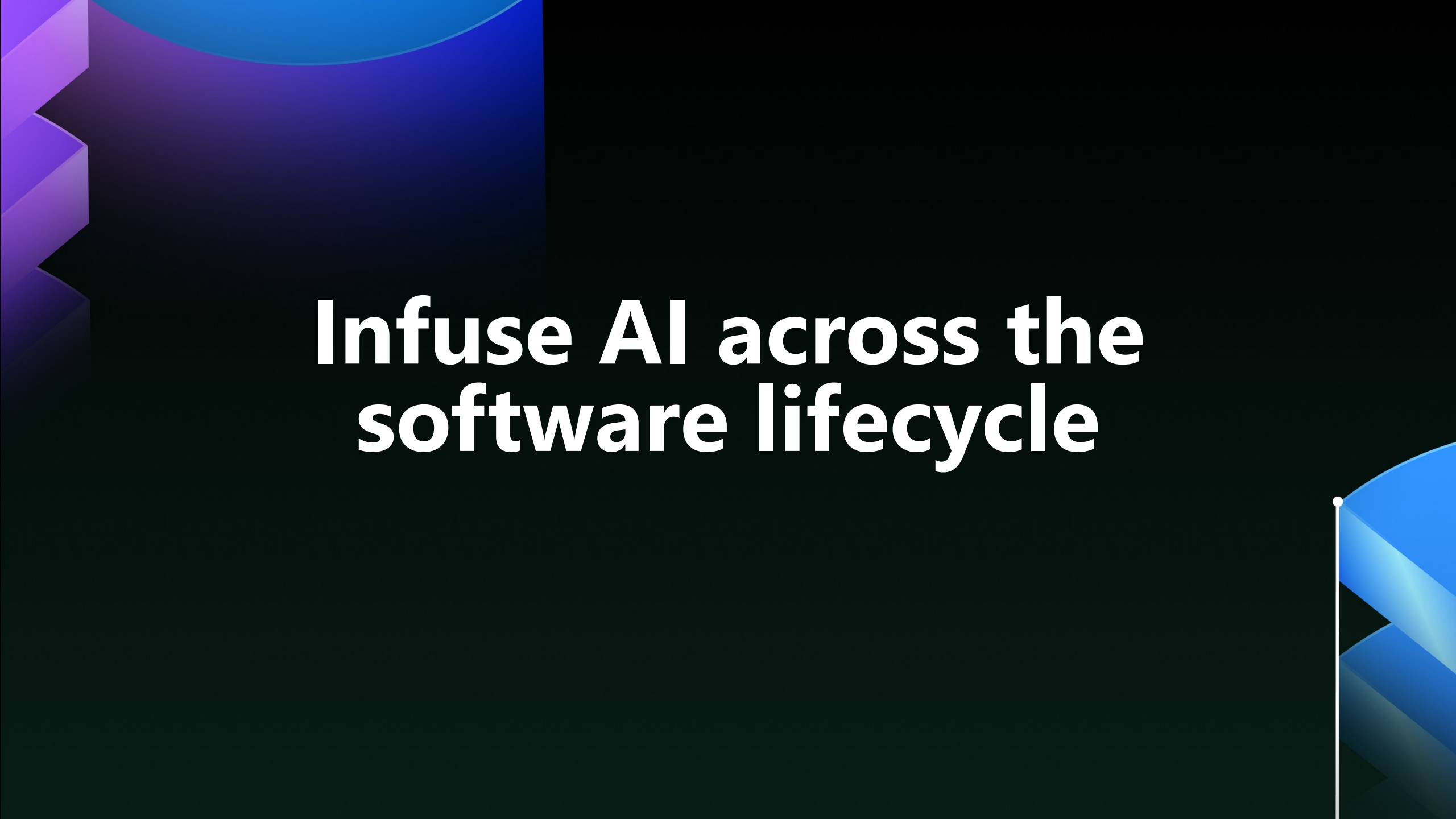
Extensibility



Marketplace



MCP



Infuse AI across the software lifecycle

Unlock the power of agents at scale across the entire software lifecycle



Establish an AI-powered development lifecycle

Support developers across more tasks by infusing AI across the entire software lifecycle



Enhance code quality and security from your agents

Get more maintainable, reliable code from your agents and enhance organization-wide visibility



Manage, monitor, steer, and review agent tasks in one place

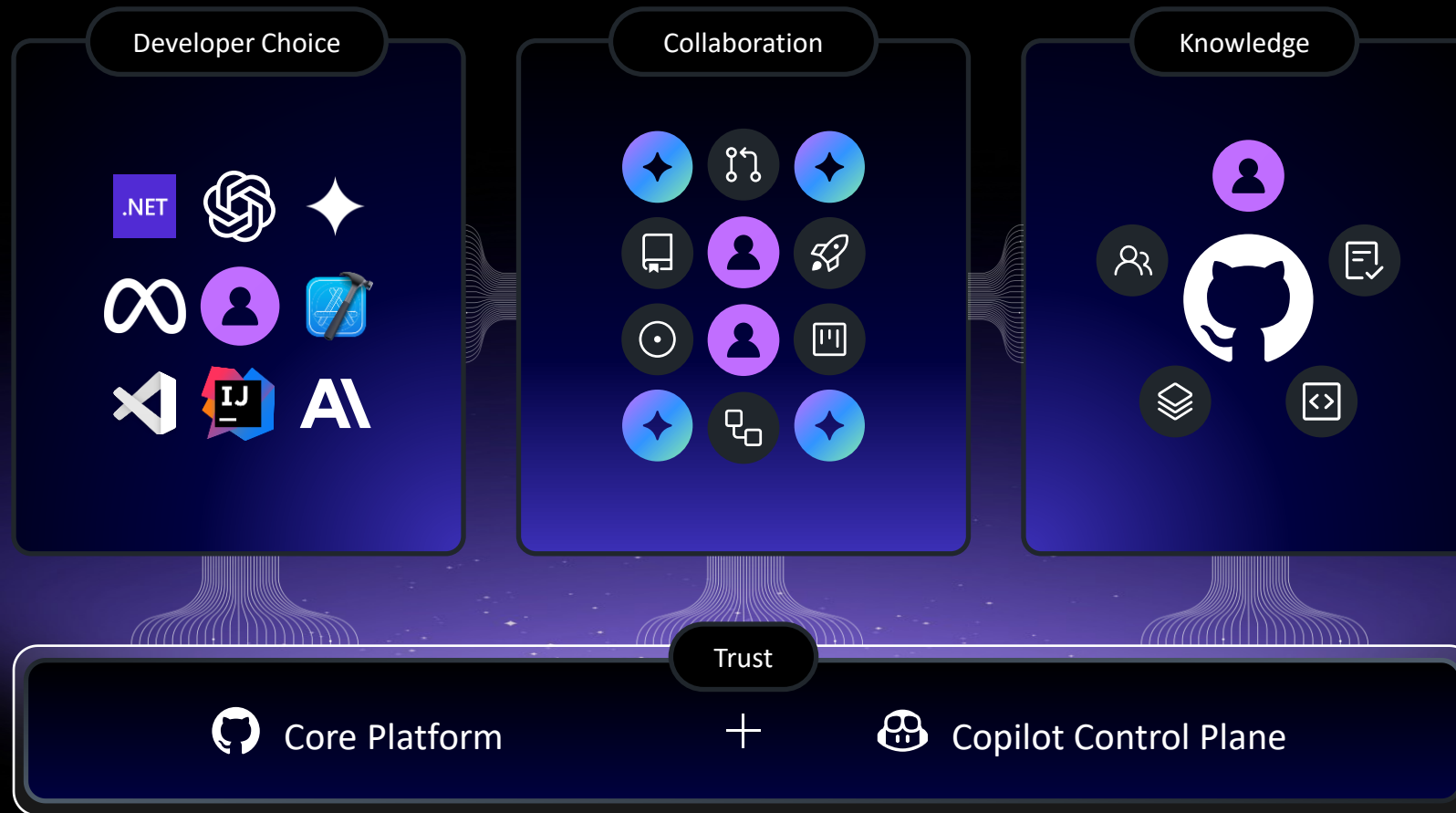
Orchestrate your agents and tasks with an agent mission control directly in your GitHub dashboard



Scale innovation with enterprise-grade controls

Give enterprise admins the visibility and governance to confidently manage your agents at scale

Agentic DevOps era



The AI-powered developer platform

Scale and governance



Plan

Issues

Projects

Spaces

/planning

Teams + Slack

Boards



Code

Editor/IDEs

Copilot CLI

/coding

Mission control

Custom agents



Verify

Pull Requests

/code-review

/secure + /auto-fix

/testing

Code quality



Deploy

Actions (CI/CD)

Releases

Apps

AI Workflows

Custom agents

Pipelines



Operate

Metrics/ROI

Spark Runtime

SRE Agent

MCP Integrations



Agent control plane



New agents available across the software lifecycle



Coding agent

Assign new tasks to Copilot from anywhere—now with security and quality built in



Code review

Have Copilot review code, then hand updates to coding agent for implementation



Code quality

Get more maintainable, reliable code and test coverage across every enterprise repository



App modernization

Modernize Java and .NET apps in a fraction of the time



SRE Agent

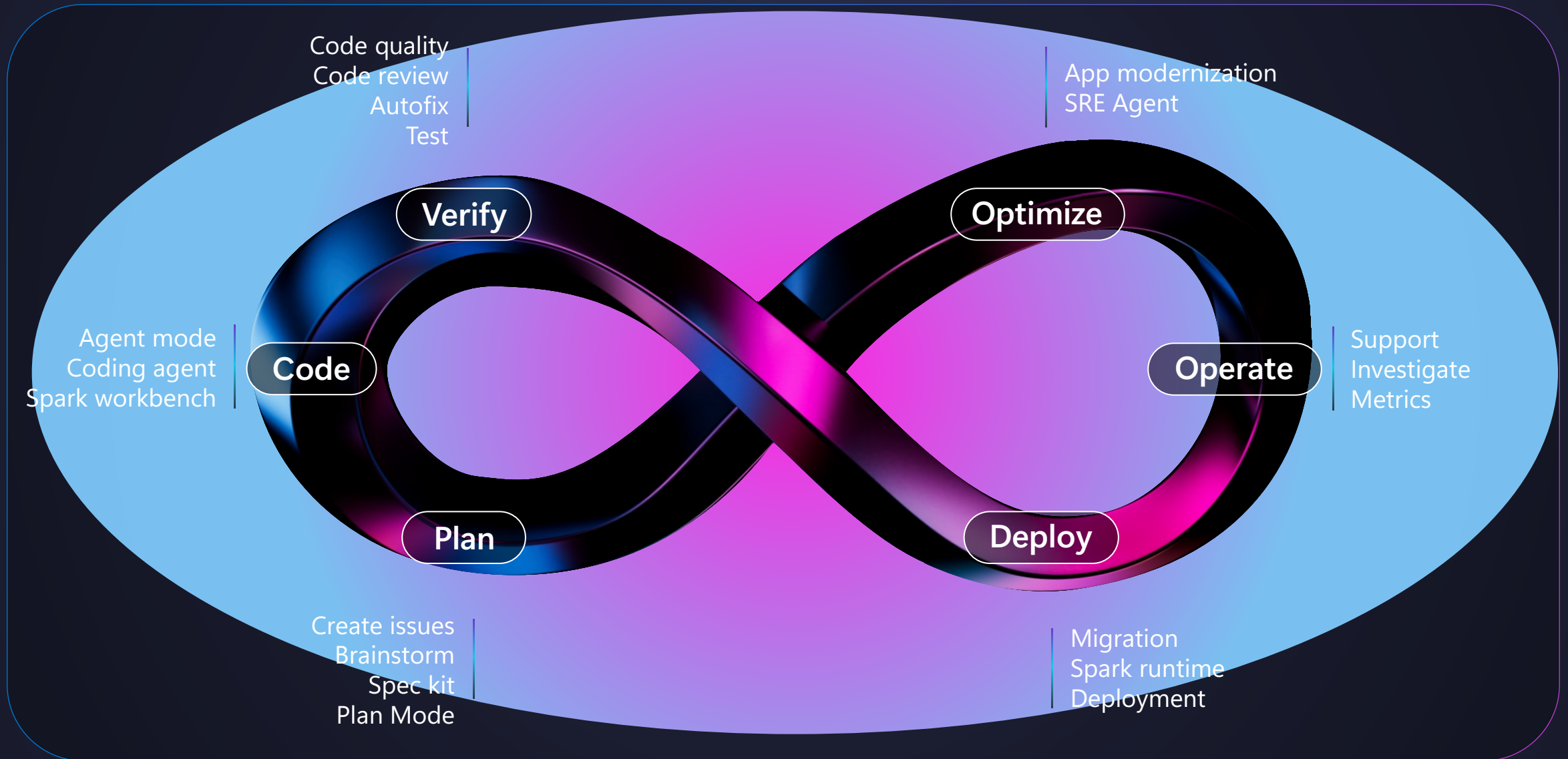
Automate operations, reduce toil, and accelerate innovation



Custom agents

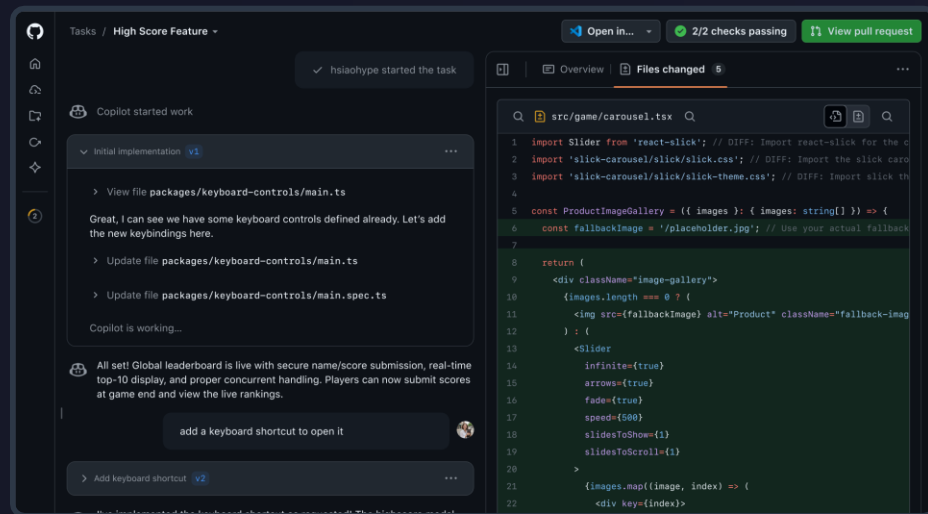
Reduce repetitive prompting by creating specialized agents to target recurring tasks

Agents across the entire software lifecycle



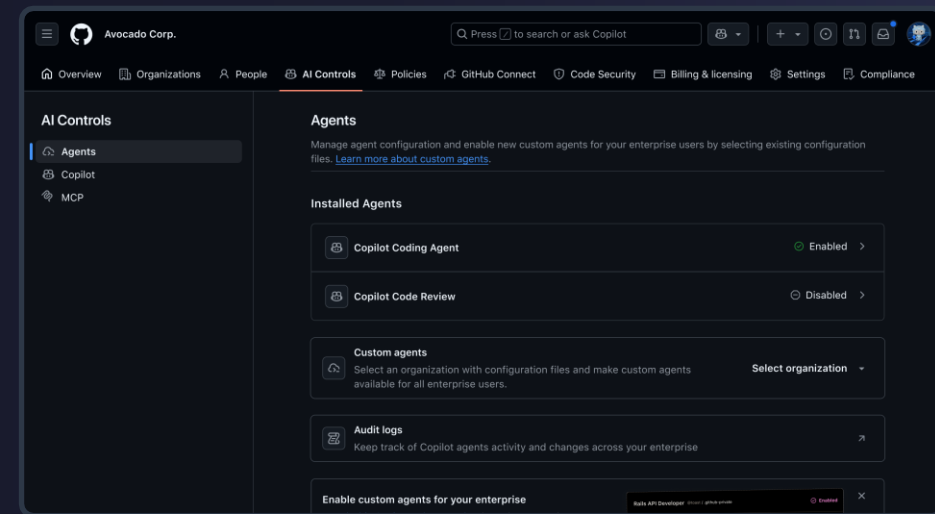
Scale AI through Agent HQ

Developer Mission Control



Jump between tasks, see what's changed, track Copilot's progress, and step in with course-corrections as needed—all in one place.

Admin control plane



Govern AI—all in one simple place—and get comprehensive visibility into Copilot adoption value, and organizational impact.

←

→

tailspin-toys

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CHAT

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Build with agent mode

AI responses may be inaccurate.

📎 Add Context...

Add context (#), extensions (@), commands (/)

Agent Claude Sonnet 4.5

🔧

🔄

➤

<

🔗 accessibility-check

🔍

⊗ 0

⚠️ 0

🔍

👤

🔔

AGENTIC DEVOPS WITH GITHUB COPILOT



What is new and upcoming

Coding Agent

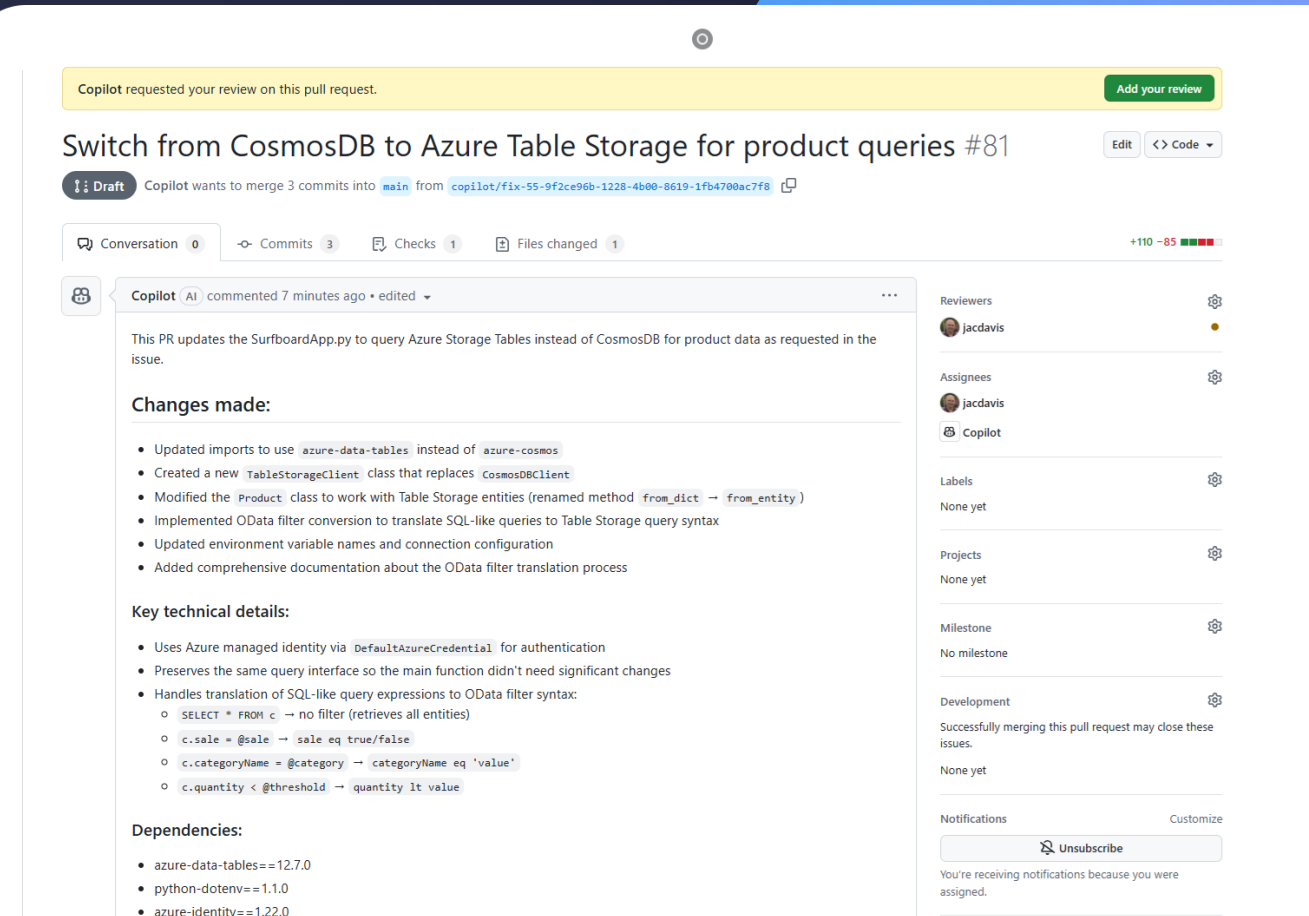
Agentic capability in GitHub to enhance team productivity like an AI teammate

Assign issues to GitHub Copilot and get back pull requests, automatically validated with tests and linters

Stay in control by working with Copilot and iterating through pull request reviews

Tailor to your needs by customizing its dev environment, adding custom instructions, and MCP servers

Proactively performs security and quality analysis of code that it creates



Agentic Workflows

Towards

Natural-Language

Programming for

GitHub Actions

[GitHub Next | Agentic Workflows](#)

```
---
on:
  push:
    branches: [main]

permissions:
  contents: read
  pull-requests: read

safe-outputs:
  create-pull-request:

tools:
  edit:
  web-fetch:
---

# Documentation Updater

You are a technical writer. Your job is to make the documentation in the repository ${github.repository}
}} _excellent_.

Steps:

1. Analyze repository changes. On every push to the main branch, examine the diff to identify
changed/added/removed entities.
2. Review existing documentation for accuracy and completeness. Identify documentation gaps including
missing or outdated sections.
3. Update documentation as necessary.
4. Create a pull request with a clear description of the changes.
```

Mental model

Your intent (Markdown)



gh-aw compiler (CLI)



Hardened GitHub Actions workflow



AI agent runs INSIDE Actions



Safe outputs + human review

Learning Resources

<https://devblogs.microsoft.com/all-things-azure/getting-started-with-agentic-devops-part-1-foundations/>

<https://azure.microsoft.com/en-us/blog/agentic-devops-evolving-software-development-with-github-copilot-and-microsoft-azure/>

<https://learn.github.com/courses>

<https://learn.microsoft.com/en-us/training/browse/?products=github>

Hands-on Exercise

1. [skills/getting-started-with-github-copilot](#): Learn, build, debug, and ship faster than ever with your favorite AI pair programmer.
2. [skills/code-with-codespaces](#): Develop code using GitHub Codespaces and Visual Studio Code!
3. [skills/modernize-your-legacy-code-with-github-copilot](#): Modernize a school's legacy accounting system from Cobol to Node.js with the help of GitHub Copilot.

Hands-on Exercise

4. skills/secure-repository-supply-chain: Secure your supply chain, understand dependencies in your environment, know about vulnerabilities in those dependencies and patch them

5. skills/secure-repository-supply-chain: Secure your supply chain, understand dependencies in your environment, know about vulnerabilities in those dependencies and patch them

TAHNC YOU